

exists some  $x \in \mathbb{Z}^n$  with  $Q(x) = a$ . This theorem has purely number-theoretic proofs (see the survey by Schulze-Pillot [335]).

In fact Ellenberg and Venkatesh proved in [83] a different theorem that currently does not have a purely number-theoretic proof. They considered the problem of representing a quadratic form by another quadratic form: If  $Q$  is an integral positive definite<sup>(6)</sup> quadratic form in  $n$  variables and  $Q'$  is another such form in  $m < n$  variables, then one can ask whether there is a subgroup  $\Lambda \leq \mathbb{Z}^n$  generated by  $m$  elements such that when  $Q$  is restricted to  $\Lambda$  the resulting form is isomorphic to  $Q'$ . This question has, for instance, been studied by Gauss in the case of  $m = 2$  and  $n = 3$  in the *Disquisitiones Arithmeticae* [111]. As before, there can be congruence obstructions to this problem, which are best phrased in terms of  $p$ -adic numbers. Roughly speaking, Ellenberg and Venkatesh show that for a given integral definite quadratic form  $Q$  in  $n$  variables, every integral definite quadratic form  $Q'$  in  $m \leq n - 5$  variables<sup>(7)</sup> that does not have small image values can be represented by  $Q$ , unless there is a congruence obstruction. The assumption that the quadratic form  $Q'$  does not have small image means that  $\min_{x \in \mathbb{Z}^m \setminus \{0\}} Q'(x)$  should be bigger than some constant that depends on  $Q$ .

The ergodic theory used in [83] is related to Raghunathan's conjecture mentioned in Sect. 1.4 and discussed again in Sect. 1.7 below, and is the result of work by many people, including Margulis, Mozes, Ratner, Shah, and Tomanov.

## 1.7 Dynamics on Homogeneous Spaces

Let  $G \leq \mathrm{SL}_n(\mathbb{R})$  be a closed linear group over the reals (or over a local field; see Sect. 9.3 for a precise definition), let  $\Gamma < G$  be a discrete subgroup<sup>(8)</sup>, and let  $H < G$  be a closed subgroup. For example, the case  $G = \mathrm{SL}_3(\mathbb{R})$  and  $\Gamma = \mathrm{SL}_3(\mathbb{Z})$  arises in Sect. 1.4 with  $H = \mathrm{SO}(2, 1)$ , and arises in Sect. 1.5 with  $H = A$ . Dynamical properties of the action of right multiplication by elements of  $H$  on the homogeneous space  $X = \Gamma \backslash G$  is important for numerous problems<sup>(9)</sup>. Indeed, all the results in Sects. 1.4–1.6 may be proved by studying concrete instances of such systems. We do not want to go into the details here, but simply mention a few highlights of the theory.

There are many important and general results on the ergodicity and mixing behavior of natural measures on such quotients (see Chap. 2 for the definitions). These results (introduced in Chaps. 9 and 11) are interesting in their own right, but have also found applications to the problem of counting integer (and, more recently, rational) points on groups (or certain other varieties). The first instance of this can be found in Margulis's thesis [252], where this approach is used to find the asymptotics for the number of closed geodesics on compact manifolds of negative curvature. Independently, Eskin and McMullen [86] found the same method and applied it to a counting problem in